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Objective

I am seeking a lead development role, with a strong focus on User Interface development, ideally for consumer facing products.

Technical Skills

- Web development using HTML, CSS, Javascript, JSON
- AJAX and rich web user interfaces
- User Interface development with Ext.js, Cocoa/Cocoa Touch, Win32, COM/ActiveX
- Server side development for web services in PHP, Java
- iPhone and Mac application development
- Unicode and Internationalization issues
- HCI, User Interface Design, UI Accessibility
- TCP/IP networking, HTTP network streaming
- Streaming audio and video, computer vision and computer graphics
- Object-Oriented design and development in C++, Java, C#
- C++, Objective-C, Javascript, PHP, Java, C#, Python

Leadership Skills

- Technical lead experience on multiple projects
- Design leadership and collaboration with User Experience staff and Product Management
- Experienced technical liaison to external platform vendors to evaluate and provide feedback for platform updates and enhancements
- Mentor and advisor to other developers

Education

Georgia Institute of Technology

M.S. in Computer Science, completed Dec. 1999

B.S. in Computer Science, completed June 1996 (Summa Cum Laude)

Experience

Infor Global Solutions (Senior Software Engineer) – July 2009 to April 2010

Development

- Extended the ClearUX user interface framework based on Ext.js, which is used by multiple Infor product teams to deliver their applications (Javascript)
- Extensive CSS and Javascript development to develop extensions to the Ext Grid, Tree, and other widgets
- Extended the declarative UI specification to support event argument lookups (Javascript, JSON, XML, XSD, Java)
- Servlet support for additional developer commands for the ClearUX Framework (Java, Servlets, Tomcat, Maven, Spring)

Leadership

- Spearheaded effort to expand unit testing coverage of ClearUX widgets
- Coordinated significant fix to flaws in the underlying JSON processing
- Provided forum support for ClearUX users to resolve developer difficulties with the framework

User Experience

- Generated widget designs and standards with UX team members to establish best practice UI patterns for Infor Applications
- Coordinated with product team members to resolve usability concerns with particular widgets and use cases

Yahoo! (Senior Technical Yahoo!) - May 2005 to February 2009

Development

- Senior Engineer for the Yahoo! Toolbar from version 6 through version 8
- Internet Explorer client features for the Yahoo! Toolbar (ATL, C++, Javascript)
- Architected the Toolbar 7 user layout model and a corresponding Customization API, accessible via Javascript
- “My Notifications” feature for Toolbar 8 (C++, Win32, ATL, GDI)
- Prototyped Toolbar tabbed browsing add on for IE 6 (C++, Win32, ATL)
- RSS feed button client UI and web service for Toolbar 7+ (C++, Win32, PHP)
- Internal JSON library for Toolbar data transfer (C++)
- Server side support for dynamic Toolbar buttons (PHP, some Perl, C++)
- Browser version and distribution deal redirector for <http://toolbar.yahoo.com> (PHP)
- Server side data source aggregation for Toolbar integration with Yahoo! Updates for the “My Notifications” feature (PHP)
- Extensions for internal content management tools for Toolbar button creation and editing (PHP)

Leadership

- Primary Yahoo! Engineering contact for Internet Explorer 7 and 8 add on compatibility
- Investigated compatibility issues with pre-release builds of IE7 and IE8
- Provided feedback to Microsoft regarding ActiveX addons, including Protected Mode issues
- Provided guidance to other engineering groups regarding best practices for working with IE7, IE8 and Vista Protected Mode
- Acted as primary engineering lead for the Toolbar 7.1 US release, coordinating the team to meet our release objectives

User Experience

- Extensive internationalization client and server support for over 30 locales (C++, PHP, Unicode, I18N)
- Accessibility support for screen readers and related assistive technologies, including work with Microsoft Active Accessibility (MSAA)

Bellsouth (Senior Software Engineer) - August 2004 to May 2005

Development

- User Interface controls, using both Windows custom control interfaces, as well as custom controls using the Qt framework for Bellsouth Applications
- Cross-platform deployment framework components targeting Windows and MacOS X
- Developed an ECMA script (Javascript) scripting sandbox layer for application development
- C++ and the Standard Template Library, Qt, ATL/COM, HTML and Javascript
- Designed and prototyped peer to peer Instant Messaging services for Bellsouth Messenger

Leadership

- Member of Architecture Team focused on a next generation user interface services
- Defined coding standards, technology selections, revision control best practices, and software engineering methodology, such as test driven development
- Designed user interface architecture for the Bellsouth Personal Desktop initiative, including a model-view-controller design for use in Bellsouth applications
- Authored design documentation for the BPD UI, and review design documents for related core services, such as user sign-in services, application logging, and system health monitoring

User Experience

- Iteratively developed the Bellsouth Messenger User Interface using recorded usability lab sessions
- Collaborated with the Creative Services team to provide effective User Interface development tools and workflow to empower designers

BBC Technology (User Interface Team Lead) - July 2003 to July 2004

Development

- Designed and implemented the Screening Support System [Triple S] for ESPN for use in their broadcast workflow for broadcast television programming
- Implemented the Triple S user interface for video metadata annotation using COM, ATL, WTL and Java middleware using XML based messaging
- Integrated Quantel Digital video solutions and QView Lite for video playback

Leadership

- Led team of two developers for user interface implementation of the Triple S user interface
- Led Agile development practices and iterative development
- Architected the UI framework and middleware integration interfaces, developed schedule estimates and performed ongoing project status reporting
- Negotiated interface functionality and design with senior ESPN management and their user populace

User Experience

- Collaborated with requirements analysts to document ESPN legacy workflow for the highlight screening process
- Interacted with ESPN management to determine workflow and cultural constraints
- Worked with BBCT User Experience staff to create application wireframe designs for the Triple S user interface
- Provided prototype UI to select user populace for feedback and revision

Bellsouth.Net (Windows Application Consultant) - May 2002 to July 2003
Development

- Email and Instant Messaging applications for BellSouth.Net customers
- COM components architecture using C++, ATL, and Visual Studio.NET
- Use of custom drawn controls in GDI and application skinning libraries
- MSN Messenger protocol plug-in, XMPP (Jabber) Messaging (TCP/IP, Jabber, XMPP, XML, SAX, C++)

Incanta, Inc. (Client Software Developer) - Jan. 2000 - April 2002
Development

- Created version 1.0 of the client application for Incanta Music, an internet radio service for Broadband Internet users as lead UI developer
- Evolved client application to version 2.0 as the sole front end developer
- Created versions 3 and 4 of the Incanta Music Player as a member of a team of four developers in C++ using MFC, DirectShow and various COM components
- Created the Service Navigator product, a Sidebar add-on for Internet Explorer as an alternative user interface and launch point for Incanta services
- Created an ActiveX wrapper for the core Incanta Music engine for embedding in the Service Navigator (C++, COM and ATL)

Leadership

- Developed Incanta Music 1 and 2 for early stage funding -- resulted in \$15 million in venture capital
- Provided initial support knowledge base for Incanta Music technical support
- Interacted with Marketing and Sales to refine products to satisfy customer needs
- Led product design revisions that led to fewer technical support incidents and higher application reliability based on feedback from customer reports and technical support
- Sole developer for the Service Navigator product, taking it from initial concept to headlining product

User Experience

- Designed the interface for the Incanta Music player based on proven UI principles and guidelines with four member user interface team
- Performed iterative User Interface design using usability lab evaluation; final usability testing rounds showed our changes had addressed our top usability problems
- Acted as user advocate in design meetings to balance usability with marketing proposals

Interactive Media Technology Center (Graduate Research Assistant) - June 1999 - Dec. 1999

- Developed a gesture recognition engine for use with a novel input device called the Haptic Lens (<http://www.imtc.gatech.edu/projects/technology/haptic.html>) to provide recognition of trained gesture alphabets via Dynamic Time Warping
- Optimized algorithm for connected components determination using dynamic programming techniques for use with DirectDraw and DirectShow in Windows 98

Nortel Networks (Member of Scientific Staff) - June 1996 - September 1998

- Designed and developed user interfaces for various Network Management software for Nortel networking equipment in C++ on a Unix platform
- Developed embedded command line front end for the AccessNode Express network element in C for the PSOS+ Embedded Operating System

Personal Projects

- Developed the iPhone task management application DulyNoted -- available in the iTunes App Store
- Worked with the Computational Perception Lab (<http://www.cc.gatech.edu/cpl>) to develop software to generate voxel-bounding volumes using calibrated camera views
- Developed an After Effects plug-in (C++) for special effects
- Examined the current state of Ubiquitous Computing in the context of audio and auditory interfaces (<http://www.mindspring.com/~iwc/cs8503/main.html>)
- Designed and coded front-end interface to the POLKA algorithm animation system (called SAMBA) for use in algorithm visualization and visual debugging [UNIX, C++] (<http://www.cc.gatech.edu/gvu/softviz/algoanim/samba.html>)
- Designed prototype user interface for mobile Emergency Medical Information Access, called SubDerma (http://www.cc.gatech.edu/classes/cs6751_98_fall/projects/team2/)

Known Languages and Environments

- C++/C (Expert), COM and ATL(Expert), Objective C and Cocoa, Cocoa Touch, C# and .NET, Microsoft Foundation Classes (MFC), OpenGL, DirectX (DirectDraw, DirectShow, DirectSound), Java and Swing, Common Lisp, HTML, JavaScript, PHP 5
- Source control: CVS, Subversion, Perforce, ClearCase, SourceSafe, Git
- Windows XP/Vista/Win7, MacOS X, Linux, FreeBSD, HP-UX, Solaris/SunOS